

System Architecture Document (SAD)

Project: LearnLens AI

1. Executive Architectural Summary

LearnLens AI is architected as a **local-first, multi-agent educational intelligence runtime**. The system decouples visual and auditory observation from pedagogical delivery through a strictly partitioned, session-scoped retrieval-augmented generation (RAG) architecture.

Core Architectural Axioms:

- User Authority:** Observation is explicitly granted via browser permission and can be interrupted or paused at any millisecond.
- Session Isolation:** Knowledge namespaces remain strictly isolated per learning session, preventing cross-domain hallucinations.
- Multimodal Grounding:** Every explanation produced by the AI Teacher is anchored to verified timestamps, speech quotes, and captured slide images.
- Local Sovereignty:** Computation, storage, and indexing run locally on the client machine.

2. High-Level System Architecture

```
graph TD
    User([Human Learner]) -->|Explicit Authorization| Browser[Chrome Browser Tab / Window]
    Browser -->|MediaStream Video + Audio| ClientApp[LearnLens Web Studio / React]

    subgraph LocalWorkstationBoundary [Local Workstation Boundary]
        ClientApp -->|REST API & WebSockets| Backend[FastAPI Backend Server]

        subgraph MultimodalObservationEngine [Multimodal Observation Engine]
            Backend --> CaptureSvc[Capture & Buffer Service]
            CaptureSvc --> Whisper[faster-whisper Audio Engine]
            CaptureSvc --> DHash[dHash Perceptual Deduplicator]
            DHash --> OCR[OCR & Text Extraction Engine]
            DHash --> VLM[VLM & Computer Vision Classifier]
        end

        subgraph KnowledgeRAGEngine [Knowledge & RAG Engine]
            Whisper --> KnowledgeSvc[Knowledge Extraction Agent]
            OCR --> KnowledgeSvc
            VLM --> KnowledgeSvc
            KnowledgeSvc --> SessionDB[(SQLite Metadata DB)]
            KnowledgeSvc --> VectorStore[(Session Vector Index)]
        end

        subgraph PedagogicalDocumentAgents [Pedagogical & Document Agents]
            VectorStore --> RAGRetriever[Session-Scoped RAG]
            RAGRetriever --> AITeacher[AI Teacher Agent]
            KnowledgeSvc --> PPTAgent[presentation-pptx Presentation Engine]
            KnowledgeSvc --> PDFAgent[ReportLab Document Engine]
            KnowledgeSvc --> QuizAgent[Assessment Prep Engine]
        end

        AITeacher -->|Timestamp & Slide Evidence| ClientApp
        PPTAgent -->|Downloadable .pptx| User
    end
```

3. Subsystem Breakdown

3.1 Browser Capture Layer (`MediaStream` & Chrome Extension)

- Utilizes `navigator.mediaDevices.getDisplayMedia({ video: true, audio: true })`.
- Provides explicit choice of Chrome Tab, Window, or Entire Screen.
- An offscreen canvas captures frames at 3000ms intervals, converting pixels to optimized JPEG buffers.
- An optional Manifest V3 extension injects a side-panel companion directly into the browser workflow.

3.2 Multimodal Observation Pipeline

1. Perceptual Difference Hashing (dHash):

- Resizes incoming frame to 9×8 grayscale.
- Evaluates horizontal gradient sign across adjacent pixels, yielding a 64-bit fingerprint.
- Calculates Hamming distance against previously recorded frame.
- If distance < 8 (threshold), the frame is dropped as stationary or redundant, saving 85%+ CPU/GPU resources.

2. Vision Language Modeling (VLM):

- Categorizes meaningful frames into: `TITLE`, `SLIDE`, `DIAGRAM`, `CODE`, `TABLE`, `DEFINITION`, `DEMO`.
- Extracts semantic visual descriptions and calculates educational importance scores (0.0 to 1.0).

3. Optical Character Recognition (OCR):

- Extracts slide headings, bullets, technical parameters, and diagram labels.

3.3 Session-Scoped Knowledge & RAG

- **Partitioning:** Each session creates a dedicated database scope and directory: `data/sessions//`.
- **Hybrid Retrieval:** Combines subword n-gram vector embeddings (cosine similarity) with BM25 keyword matching.
- **Evidence Formatting:** Prepares grounded citation cards containing `{ timestamp, frame\_id, thumbnail\_url, quote, concept\_name }`.

3.4 Personal AI Teacher Engine

- Offers 12 teaching personas: Simple, Detailed, Exam Focus, Quick Revision, Example, Teach From Scratch, Active Recall, Flashcards, Practice Quiz, Weak Areas, Compare Concepts, Ask Anything.
- Strict anti-hallucination protocol: explicitly demarcates verified captured material from general background knowledge.

3.5 Automated Document & Presentation Agents

- **PPT Generator (`python-pptx`):** Generates 16:9 widescreen presentation decks incorporating course titles, agendas, definitions, architecture steps, real-world analogies, exam traps, and embedded screenshot images.
- **PDF Generator (`reportlab`):** Compiles PDF A (Visual Study Pack) and PDF B (AI Teaching Report with executive summary, tables, and self-check questions).

4. Agent State Machine & Human Control

```
stateDiagram-v2
[*] --> IDLE
IDLE --> OBSERVING: User clicks 'Start Learning'
OBSERVING --> PROCESSING: New frame or audio chunk
PROCESSING --> OBSERVING: Frame deduplicated / analyzed

OBSERVING --> PAUSED: User clicks 'Pause Agent'
PROCESSING --> PAUSED: User pauses
PAUSED --> OBSERVING: User clicks 'Resume Agent'

OBSERVING --> INTERRUPTED: User sends new command
PAUSED --> INTERRUPTED: User asks question
INTERRUPTED --> OBSERVING: Command executed

OBSERVING --> COMPLETED: User clicks 'Stop'
PAUSED --> COMPLETED: User clicks 'Stop'
COMPLETED --> [*]
```

5. Security & Isolation Matrix

| Subsystem | Security Enforcement |

| :--- | :--- |

| Screen Capture | Native browser prompt; zero silent execution; user can terminate anytime. |

| Credential Protection | No access to Chrome cookies, local storage, password vaults, or network headers. |

| Data Persistence | SQLite database and session files stored strictly in local repository directory. |

| Assessment Integrity | Formative practice only; no automated submission of graded institutional exams. |